



December 2024

CY1006-1 – CHEMISTRY OF ENGINEERING MATERIALS

(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Methanol is one of the most electroactive organic fuels in the low-temperature range because**
A) It has a low carbon content
B) It has a high carbon content
C) It has readily oxidizable OH⁻ group
D) A and C are correct
- The anode current collector in the Li-ion battery is made up of**
A) Iron foil
B) Carbon foil
C) Copper foil
D) Aluminium foil
- Polymer Electrolyte Membrane (PEM) Fuel cell is also called as.....**
A) Physical Electrolyte Membrane Fuel cell
B) Electron Exchange Membrane Fuel cell
C) Proton Exchange Membrane Fuel cell
D) Proton Electron Membrane Fuel Cell
- In the flow battery, the active chemical components will be taken with**
A) Anode
B) Cathode
C) Membrane
D) Electrolyte
- Corrosion between the dissimilar metals is called as.....corrosion.**
A) Galvanic
B) Dry
C) Oxidation
D) Concentration cell
- In an electroless plating, the surface of the substrate will be activated by treating.....**
A) SnCl₂ and PdCl₂
B) Alkali
C) Alcohol
D) Water
- In anodic coatings, the coating metal possess reduction potential than base metal.**
A) Higher
B) Medium
C) Lower
D) Equal
- The Full form of PCB is.....**
A) Polished Circuit Board
B) Printed Circuit Board
C) Polymer Circuit Board
D) Printed Circular Board
- A working concentration cell is made of two half cells having**
A) Identical electrodes
B) Identical electrolytes
C) Identical reactive ion concentrations
D) Only A & B
- Calomel electrode is an example for.....**
A) Metal-metal ion electrode
B) Ion selective electrode
C) Metal- metal salt ion electrode
D) Primary reference electrode
- pH of the solution can be determined using the following electrode combination**
A) Glass-calomel electrode
B) Glass-Zinc electrode
C) Platinum-Calomel electrode
D) Zinc- Calomel electrode
- Emf of the concentration cell Ag(s)|Ag⁺(0.01M)||Ag⁺(0.1M)|Ag at 298 °K is**
A) 0.0591V
B) 0.02955V
C) -0.0591V
D) -0.02955V

13. In an Electrochemical sensor, the active component that converts chemical reactions into measurable electrical signals is called as.....
 A) Receptor B) Amplifier
 C) Detector D) Transducer
14. Conductometry can be employed to determine equivalence point in
 A) Acid-base titrations B) Redox titrations
 C) Complexometric titrations D) Precipitation titrations
15. Butyl rubbers are.....
 A) Homopolymers of acrylates B) Condensation polymers of dimethyl silanol
 C) Copolymers of isobutylene and isoprene D) Condensation polymers of bisphenol A and epichlorohydrine
16. Kevlar fibre is obtained by reacting *p*-phenylene diamine with.....
 A) Terephthaloyl chloride B) 2-naphthol
 C) Bisphenol A D) Dimethylchloro silane
17. Which is the preferred precursor in the sol-gel method?
 A) Metal chlorides B) Metal nitrates
 C) Metal hydroxides D) Metal alkoxides
18. Liquid crystals are usually composed of.....
 A) Coil like molecules B) Rodlike molecules
 C) Oval molecules D) Triangular molecules
19. Which of the following is NOT a source of e-waste?
 A) Plastic bottles B) Printer
 C) Fluorescent lamps D) Laptops
20. Leaching agent used in the extraction of copper from e-waste
 A) Conc. HNO_3 + Conc. HCl B) Conc. HNO_3 + Conc. H_2SO_4
 C) Conc. HNO_3 + Excess of Ammonia D) Conc. H_2SO_4 + Excess of Ammonia

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the construction and working of the Li-ion battery. Mention any two advantages of Lithium as an electrode material.	7	L*2	1	1
	b) Describe the construction and working of the methanol oxygen fuel cell. Mention any two applications of it.	7	L2	1	1
	c) What are reserve batteries? Mention any two advantages of reserve batteries.	2	L1	1	1
2.	a) Elucidate the electrochemical theory of corrosion by taking iron as an example.	8	L2	2	1
	b) Describe the galvanization and sacrificial anode method of prevention of corrosion.	8	L2	2	1
3.	a) Explain the construction and working of solar photovoltaic cell and mention any two advantages of it.	8	L2	1	1
	b) Discuss the process involved in the electroless plating of PCB.	6	L2	2	1
	c) Waterline corrosion is a type of differential aeration corrosion. Justify the statement.	2	L3	2	1
Unit – II					
4.	a) With a neatly labelled diagram explain the construction and working of the Calomel electrode.	6	L2	3	1
	b) Explain the determination of pH using a glass electrode. Mention any two advantages of the glass electrode.	6	L2	3	1
	c) A Concentration cell was constructed by immersing two copper electrodes in 0.065 M and 0.056 M CuSO_4 solution. Write the cell scheme and calculate EMF of the cell.	4	L3	3	2

5. a)	In a sample of a polymer, 27% of molecules have a molecular mass of 17500g/mol, 36% of molecules have a molecular mass of 27000g/mol, and the remaining molecules have a molecular mass of 36500g/mol, calculate the number average and weight average molecular mass of the polymer.	6	L3	4	2
b)	Write synthesis, properties and applications of (i) Polystyrene	8	L2	4	1
c)	(ii) Polymethyl methacrylate (PMMA)	2	L1	4	1
	What are polymer composites?				
6. a)	Derive an expression for glass electrode potential (Eg).	8	L2	3	1
b)	What is specific conductance? Explain the application of conductometric titration in the estimation of strong acid.	4	L2	4	1
c)	What are sensors? Explain the construction of electrochemical sensors.	4	L1	4	1
Unit – III					
7. a)	Explain the different classes of nanomaterials along with the example.	6	L2	5	1
b)	Explain the application of Liquid crystals in LCD.	7	L2	5	1
c)	Write any three effects of e-waste on the environment and human health.	3	L1	5	1
8. a)	Write a note on organic materials used in the optoelectronic devices.	4	L1	5	1
b)	Explain the synthesis of nanomaterials using the CVD technique.	6	L2	5	1
c)	Discuss the extraction of gold from e-waste.	6	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
